

BMW Sales Association Rules Report

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Repository: [Association-Rules](#)

1. Data Preparation Process

The original dataset, sales_data.csv, contained BMW sales transactions over a specific period. Each record included attributes such as:

- **Transaction Date** – when the vehicle was sold
- **Model** – BMW model sold
- **Region / Dealership** – geographic location of sale
- **Financing Type** – cash or financing
- **Add-ons / Packages** – additional options purchased

Cleaning & Transformation Steps

1. Missing Values and Duplicates:

- Removed rows with missing critical information (e.g., missing model or transaction date).
- Deduplicated repeated transactions.

2. Standardization:

- Standardized model names (e.g., BMW X3, X3, X 3 → X3).
- Cleaned add-on names to unify labels (e.g., Premium Sound, Premium Audio → Premium Sound).

3. Transaction Encoding:

- Converted each sale into an **itemset**:
["Model:X3", "Region:Nairobi", "AddOn:PremiumSound", "Financing:Yes"].
- Used TransactionEncoder to transform itemsets into a binary matrix suitable for **Apriori**.

4. Filtering:

- Set a minimum support threshold (e.g., 0.05) to focus on meaningful patterns.
- Optional: grouped dates by month/quarter to capture seasonal trends.

5. Association Rule Mining:

- Applied mlxtend's apriori algorithm to find frequent itemsets.
- Generated rules using metrics: support, confidence, lift.
- Filtered for high-confidence, high-lift rules relevant to business decisions.

2. Key Findings

Antecedent(s)	Consequent(s)	Support	Confidence	Lift
Model:3Series	AddOn:PremiumSound	0.10	0.35	1.40
Financing & Region:Nairobi	Model:X3	0.08	0.55	1.30
Model:X5 & AddOn:ExtendedWarranty	Model:X5	0.07	0.70	1.50

- **3 Series → Premium Sound Add-on:** 3 Series buyers are significantly more likely to select the Premium Sound package, indicating cross-selling potential.
- **Financing + Nairobi → X3:** Customers financing their purchase in Nairobi are more likely to buy X3 models, highlighting a regional financing trend.
- **X5 → Extended Warranty:** X5 buyers frequently purchase extended warranties, suggesting targeted upsell opportunities.

3. Business / Domain Insights

Cross-Selling Opportunities

- Promote **Premium Sound packages** directly to 3 Series buyers, both online and in dealerships.
- Consider bundling X5 purchases with **Extended Warranty** to increase revenue.

Regional Marketing & Financing

- Target Nairobi region with **financing promotions** for X3 models.
- Tailor marketing campaigns to regions and customer segments based on discovered patterns.

Inventory Planning

- Ensure dealerships with high 3 Series inventory also stock add-ons like Premium Sound.
- Highlight warranty bundles prominently for X5 customers.

Strategic Recommendations

- Use association rules to inform **recommendation systems** for customers browsing BMW configurations online.
- Align promotional campaigns, inventory, and sales incentives based on frequent co-purchases and high-lift rules.

4. Conclusion

Association rule analysis on BMW sales data provides actionable insights:

- Strong cross-sell opportunities exist between vehicle models and add-ons.
- Regional financing trends can inform targeted campaigns.
- Inventory and bundle strategies can be optimized to maximize revenue.

Future improvements include:

- Incorporating **customer demographics** (age, purchase history) to refine rules.
- Time-series analysis for seasonal trends in BMW sales.
- Visualizing association rules using **network graphs** to highlight key connections.

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GitHub: <https://github.com/VICTOR21-A/Association-Rules>